

1	Course Name:
	Computer Organization and Operating Systems
2	Course Code:
	ISB16204
3	Year and Semester Offered:
	Year 1 Semester 2
4	Credit Hours Value:
	4 Credit Hours [4 SLT Credit]
5	Prerequisite(s), if applicable:
	NIL
6	Course Synopsis: This course introduces the principles of computer organization and the basics of operating system concepts. It provides students with basic understanding of hardware units (computer components, peripherals and interconnections) and some performance enhancing methods like pipelining and memory caches. This course also presents fundamental concepts related to the design and implementation of operating systems (operating system structure, process scheduling, process synchronization, memory management). Assembly language will be used in this course to demonstrate the interaction between hardware and software components of computers.
7	Course Learning Outcome: Upon completion of the course, students should be able to: 1. Discuss the main components and organization of computers and the fundamental concepts and principles of operating systems. 2. Show the interaction between the hardware and software components of a system using Assembly language. 3. Analyse the implementation of performance enhancement mechanisms in both hardware structure and organization and operating system components and services.
8	Content Outline:
	Topic
	1. Introduction to Computer Organization & Operating System
	2. Computer Hardware & Information Flow
	3. Data Representation
	4. Digital Logic
	5. Central Processing Unit
	6. Memory Architecture
	7. Input/Output & Storage
	8. Processes & Threads

	9. CPU Scheduling
	10. Process Management & Synchronization
	11. Memory Management
	12. Virtual Memory